

Teaching Quality Is Linearly Decodable

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Summary

RL for tutoring needs a per-turn reward for teaching quality. LLM-as-judge is accurate but far too slow to sit inside a training loop. We fit a ridge regression from a frozen OLMo-2-7B-Instruct’s mid-layer activations to human ratings of a tutor turn, and use it as the reward for GRPO. It costs one dot product per turn.

- **Outcome-based rewards cannot work, measurably.** Rated teaching quality correlates with the simulated student solving at $r = -0.012$; rated answer-leakage at $+0.291$. Student outcome carries one signal, and the only lever a tutor has on it is giving the answer away.
- **The probe’s claim survives a blind check.** It claimed $+0.36$ over the untrained policy on 60 held-out questions; a human rater and a panel of six models, neither able to see the scores or which policy wrote which turn, paid $+0.34$ and $+0.35$.
- **Three failures the reward curve could not show.** Turns collapsed toward brevity (12% rated too short $\rightarrow$ 29%). One arm claimed half again more than it was paid. And an arm that fixed the length problem was ranked *worse* by both automated judges while the human preferred it 29–11, because both tie leakage to length about $1.6\times$ as strongly as she does.
- **The reward model is the ceiling, not the optimiser.** Across seven arms, learning rates spanning $4\times$ converge to the same reward. Capability is untouched: -0.3 points pooled over ARC-Challenge, MMLU stem and GSM8K.

1 The Data

Released US state exam items, with dialogues generated on them by two off-the-shelf models.

Questions. 307 released multiple-choice items from public-school standardised tests in five US states — California, Pennsylvania, Massachusetts, New Jersey, Texas — published 2001–2026. Four options each with a known key. Grades 3–11, modal grade 8; roughly 40% mathematics, 32% science, 28% social studies. Chosen because the correct answer is externally given, so nothing has to be adjudicated, and because they are what school students are actually set.

Dialogues. 299 of those items generated 1200 tutor–student conversations: 3600 tutor turns and 4800 student turns, three tutor turns each. Tutor is OLMo-2-7B-Instruct, the model later trained; student is Qwen-2.5-1.5B-Instruct prompted to work the problem and err plausibly. Four tutor prompt styles in equal share — *brief*, *socratic*, *plain*, *explain* — so the corpus spans a range of teaching behaviour rather than one house style.

Reading a teaching reward out of a frozen model — three parts

1. Reward the student's answer

ABANDONED



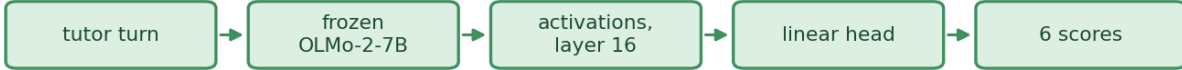
Four runs. Leaking fell every time, teaching never moved.

Quality vs solving $r = -0.012$ Leaking vs solving $r = +0.291$

Outcome carries one signal, and the only lever on it is giving the answer away.

2. Reward the turn itself

DONE



Fitted on 600 human-rated turns. One dot product per turn at training time.

targeted: 0.85 from activations against 0.36 from crude text statistics

leak hardened: answer-injection attacks fool it 3% of the time, down from 93%.

3. Train against it, and check with people

DONE



Arm A claimed +0.36, two independent panels paid +0.34 and +0.35.

Length is fixed: arm E has 100% of turns rated right-length, against base's 71%.

2x, 4x and 8x the step size all reach the same reward, so the ceiling is the reward model.

Academic ability is unchanged: -0.3 points over ARC, MMLU stem and GSM8K pooled.

Figure 1: The three parts of the project. The first is a dead end, drawn because it is why the reward reads the tutor's turn rather than the student's outcome, and because it cost four training runs to establish.

Ratings and splits. 600 tutor turns were sampled and rated by hand and by models; this is what the probe is fitted on. Questions split 250 train / 50 evaluation with no overlap. Each (question, dialogue-so-far) pair is one prompt, giving 1000 training and 200 evaluation prompts.

2 Why Not Reward the Student's Answer

Student outcome carries one signal, and answer-leakage is the only lever on it.

Four GRPO runs rewarded the tutor when a simulated student subsequently solved the problem, across two reward formulations, two corpora, and a corrected leak detector. All four taught the tutor to stop giving answers away; none improved held-out teaching.

Property of the tutor turn	Correlation with the student solving
Rated tutoring <i>quality</i>	-0.012
Rated <i>answer leakage</i>	+0.291

Table 1: The best-rated turns had the worst solve rates.

Table 1 is the cause. Outcome carries one signal — whether the student was told the answer — so penalising leakage does not add a pedagogical signal, it removes the only one present. Leak rate

	what changed	what it asked	what came back
A	first run raw mean of 4 qualities	does the reward move at all?	reward 0.83 → 1.62, but turns collapsed to 12 words
▼			
B	z-score each quality	do unequal spreads distort it?	identical to A — scalarisation was not the problem
▼			
C	hard length band, 30-58 words	can length be fixed by a band?	median 46 words, but 29% now too short: no floor
▼			
D	5th quality (correct) + linear ramps	does a ramp beat a hard edge?	crashed at step 46 (preempted before first checkpoint)
▼			
E	lr 2e-5, 200 steps band 18-40	the measured-best step size	reward 3.84, KL 0.25, 100% right length, human prefers 29-11
▼			
F	lr 4e-5	does 2× lr change the ceiling?	reward 3.88, KL 0.29 — same place in half the steps
▼			
G	lr 8e-5	does 4× lr change the ceiling?	reward 3.86, KL 0.29 — same place, and no further drift than F

**E, F and G land in the same place at 2×, 4× and 8× the step size:
the optimiser sets how fast, the reward model sets how far.**

Figure 2: Every arm: what it changed, what it asked, what came back. The arms were sequential, each designed after reading the last. A and B show the scalarisation was not the problem; C and D are about length; E, F and G vary only the learning rate.

duly fell in every run while held-out teaching never moved.

The simulated student is trapped between two failures with nothing between them. A weak student *cannot* learn: expert tutoring moved a 0.5B student’s answers by roughly 7 percentage points, which upper-bounds anything a reward computed from its accuracy could detect. A strong student *already knows the answer*: told to act confused, its chain of thought solves the problem anyway, or it drops the persona and explains like the teacher. Filtering to harder items tightens the trap, since items a weak student fails are those a strong model gets unaided. A 3B model in between was tried and abandoned.

Two reward-side remedies also failed. A dense learned quality score turned out to detect *which model wrote the turn*, and drove question-asking from 79% to 32%. Removing the leak detector’s false positives more than doubled its precision on mathematics and moved the held-out measure by 0.008.

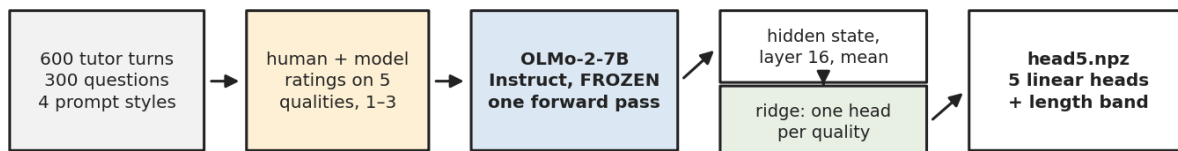
The premise of everything following: the measured quantity must admit a pedagogical channel before a reward built on it can find one. We discard the student and rate the turn.

3 The Reward Model

A ridge fit on frozen mid-layer activations reads teaching quality where surface features cannot.

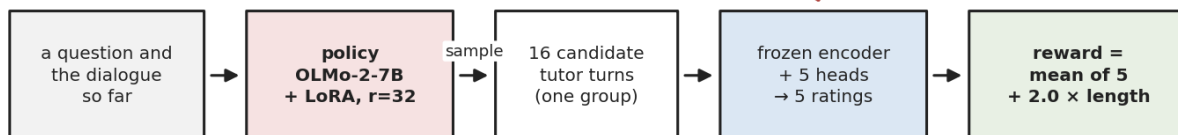
Six qualities of a turn are rated 1–3 by humans: how much it leaks the answer, how **targeted** to this student’s error, how **actionable**, how much it **elicits** the student’s reasoning, whether its **length** fits, and whether anything is **incorrect**. A ridge regression predicts each from the frozen encoder’s activations at a middle layer.

ONCE · fit the reward model against human ratings



the same frozen weights,
never updated

EVERY STEP · train the tutor with GRPO



advantage = reward − group mean → update LoRA only (KL penalty $\beta = 0.02$)

nothing outside the LoRA adapter is ever updated

Figure 3: The probe is fitted once, then called on every rollout. The encoder appears twice and is the same frozen weights both times: the policy cannot raise its own score by moving its internal representations, because the representations being read are not its own. Only the LoRA adapter is updated.

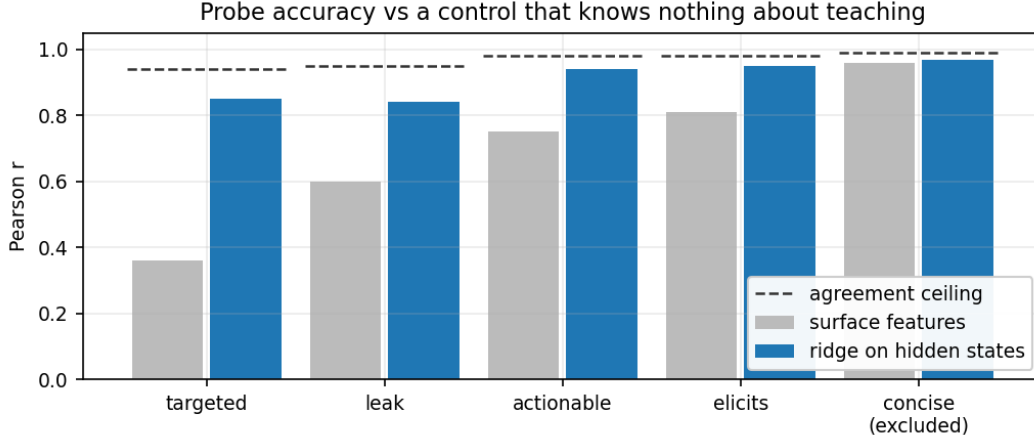


Figure 4: Probe accuracy against a control. Grey is eight surface features — word count, question marks, digit density — that encode nothing about teaching. Where grey reaches blue, the rating is a property of the text’s shape and the activations bought nothing.

The control in Figure 4 is load-bearing. **concise** is predicted at 0.96 by word count against 0.97 from activations — a word counter wearing a rubric, excluded from the reward. **targeted** is the opposite: 0.36 from surface features against **0.85** from activations. Whether a turn addresses the student’s actual mistake is nearly invisible in surface form, and the model tracks it anyway.

Ridge beat a PCA-fronted, separately tuned MLP on every quality, so the map is effectively linear. Layers 16–20 of 32 beat both ends; the final layer is occupied predicting the next token rather than summarising the turn.

Hardening against the obvious attack. Injecting the gold answer into an otherwise withholding turn fooled the first leak head 93% of the time; after adversarial training, **3%**, including on phrasings and insertion points never fitted on, with accuracy on genuine turns holding at $0.84 \rightarrow 0.82$. The fitting script refuses to emit a head missing more than 15% of held-out attacks.

4 Training Setup

GRPO on OLMo-2-7B-Instruct with LoRA, $r = 32$, on one GPU.

GRPO samples 8 candidate turns per prompt and moves the policy toward those scoring above their group mean. Only a LoRA adapter at $r = 32$ is trained; the 7B base is frozen. Throughout we report KL divergence from the initial policy (“drift”, in nats), because a reward can always be raised by moving further, so a gain is only interesting per unit of distance travelled. The two first-round arms differ only in scalarisation.

	Arm A	Arm B
Scalarisation	plain average	each score z-scored, then averaged
Hardware	1×H100	1×H200

Table 2: The two first-round arms.

Arm B exists because a plain average weights each quality by how much it happens to vary. **elicits** varies most and **targeted** least, so a plain average leans 1.5× harder on the quality surface

features already predict at 0.81 than on the one they predict at 0.36 — backwards for this project.

5 Training Results

Reward rises, generalises to held-out questions, and length falls throughout.

Reward rises $1.45 \rightarrow 1.63$ and rises as much on the 50 held-out questions as on the 250 trained on. Length falls throughout: arm A $38 \rightarrow 27$ words, arm B $43 \rightarrow 23$. Arm B’s scalar sits at zero by construction, since z-scoring centres it, so arm B must be read through its individual qualities.

Three of four rewarded qualities improve by 0.25–0.40 points. **targeted** is flat in both arms, a property of the labels rather than the training: the untrained policy already scores 2.58 of 3, and a linear fit on ratings that lopsided hedges to the middle and never predicts a full 3.

Is 1.6 high? Against the corpus the probe was fitted on, 1.60 is the 78th percentile and below the mean of the best prompt style — so the policy is scored *inside* the range the fit learned from rather than off its end. That stops holding above roughly 1.8, which arm A approaches.

6 Blind Evaluation

The probe claimed +0.36; two independent panels paid +0.34 and +0.35.

The predecessor project’s reward improved for four consecutive runs while the quantity it stood for never moved, so a rising curve is not evidence. Sixty held-out questions were answered by three policies — untrained, arm A, arm B — giving 180 turns with policy identity and scores stripped and identifiers salted. One human rated 51 turns and judged 47 pairs. Six frontier models rated all 180 after calibration on 26 of her ratings and *validation against 25 they were never shown*. Every policy answered the same questions, holding difficulty and student state fixed.

6.1 Claimed against paid

Who is measuring	Untrained	Arm A	Arm B	Gain, A	Gain, B
Model raters, 180 turns	1.32	1.65	1.71	+0.34	+0.40
Human, 51 turns	1.31	1.66	1.68	+0.35	+0.37
Reward model	1.23	1.60	1.72	+0.36	+0.49

Table 3: Claimed gain against paid gain. Absolute levels are not comparable across rows; the gain is.

Arm A is clean: claimed +0.36, paid +0.34 and +0.35. That three-way agreement is the strongest evidence the training did something real rather than exploiting the scorer.

Arm B is where a gap opens: claimed +0.49, paid +0.37 and +0.40. Mild, but it is the shape of a policy optimising the scorer rather than the quantity it represents, and it appears in the arm that optimised harder. The spread of reward across turns also collapses from 0.36 untrained to 0.18 and 0.13 — a policy settling into one shape of answer.

6.2 Which qualities moved

Disagreeing per turn and agreeing per policy are different measurements. On actionable the model raters match the human at 0.03 — chance at ranking one turn against another — yet both panels produce the same gap between policies, same direction and nearly the same magnitude

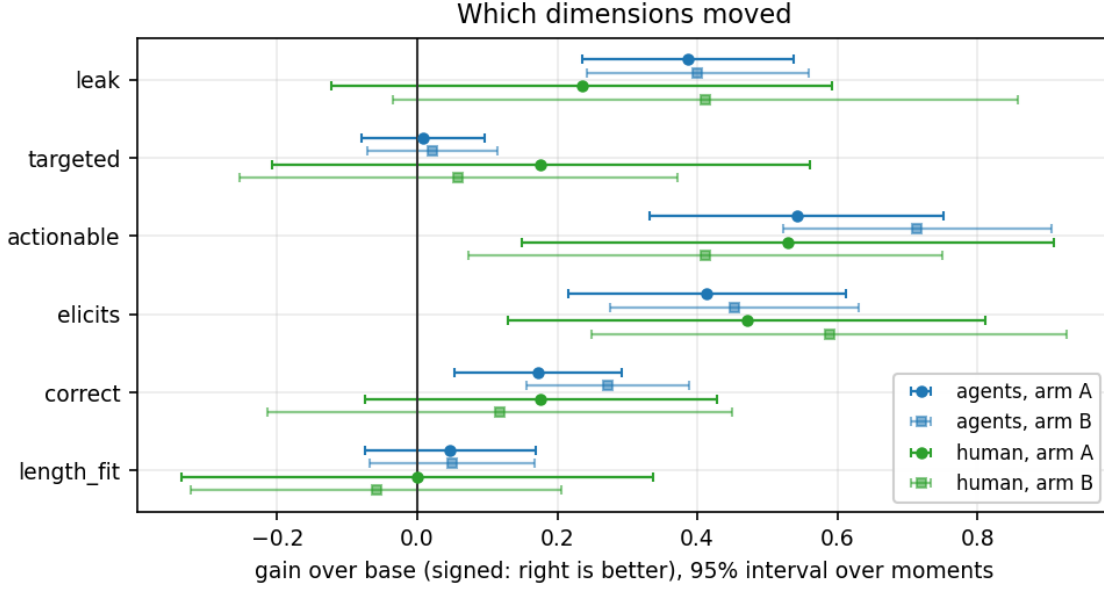


Figure 5: Gain over untrained, signed so right is always better, with 95% intervals over questions. The human’s intervals are wider throughout: 17 questions against 60.

Quality	Arm A	Arm B	Human, A / B	Raters match human, per turn
leak	+0.39 (69%)	+0.40 (72%)	+0.24 / +0.41	0.37
actionable	+0.54 (69%)	+0.71 (82%)	+0.53 / +0.41	0.03
elicits	+0.41 (64%)	+0.45 (70%)	+0.47 / +0.59	0.21
correct	+0.17 (65%)	+0.27 (73%)	+0.18 / +0.12	0.38
targeted	+0.01	+0.02	+0.18 / +0.06	0.24
length_fit	+0.05	+0.05	+0.00 / −0.06	0.26

Table 4: Bold exceeds its margin of error. Percentages are the share of questions won. The last column runs 0 (chance) to 1 (perfect).

(+0.54/ + 0.71 against +0.53/ + 0.41). Ranking noise cancels across sixty questions; a consistent preference does not. Near-zero per-turn agreement therefore disqualifies these raters for *labelling training data* and says little about *comparing policies*. Conflating the two would have discarded this evaluation’s main result.

6.3 The regression the reward curve could not show

length_fit scores 2 for right and 1 or 3 for the two ways of being wrong, so its mean hides an even split — a policy divided evenly between the two failures averages exactly ideal. The model raters put too-short at 20% and 15%, about half the true figure: the one place where trusting them alone would have hidden a real loss.

	Too short	Right	Too long
Untrained	12%	71%	18%
Arm A	29%	71%	0%
Arm B	35%	65%	0%

Table 5: Human length judgements. Training eliminated over-long turns and paid triple the too-short rate, leaving the mean flat.

Comparison	Result	Probability by chance	Agreed with reward model
Arm A vs. arm B, 17 pairs	8–9, no ties	100%	47%
Arm A vs. untrained, 30 pairs	18–10 , 2 ties	19%	68%

Table 6: Blind pairwise preferences, same question on both sides.

6.4 Head-to-head preferences

The two arms are indistinguishable to a person. Against untrained, training wins 64% of decided pairs — right direction, but 28 decided pairs cannot rule out chance where a 64% effect needs roughly 60. With the absolute ratings, which rest on far more judgements and put arm A ahead on 76% of questions, we conclude training beat the untrained policy.

The final column is more informative: the reward model *cannot* separate two trained policies (47%, scoring the two sides within 0.07) but *can* separate trained from untrained (68%). Real but coarse resolution is the honest characterisation.

6.5 Why the model ratings were not used to refit the probe

Eight features that know nothing about teaching, against six frontier models

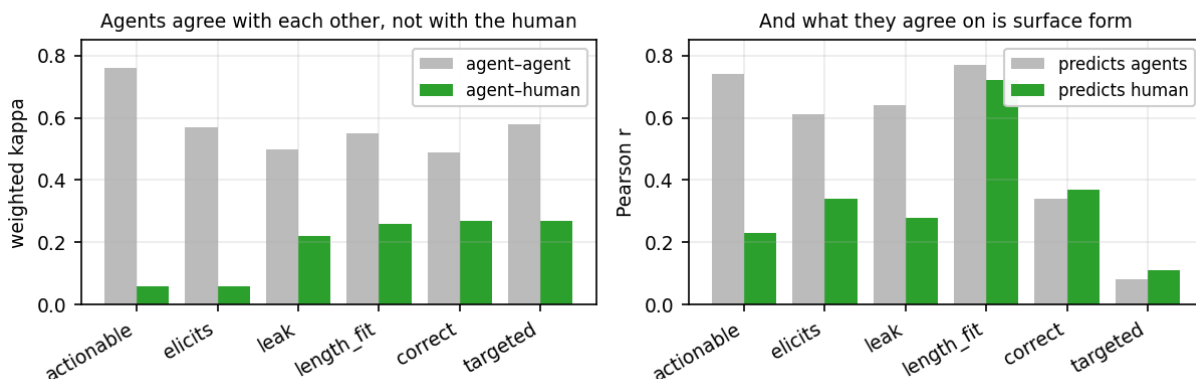


Figure 6: Left: the six raters agree with each other far more than with the human they were calibrated to. Right: what they agree on is largely surface form.

The shortcut is to treat six frontier models as a cheap annotation panel. Figure 6 is why we did not: on *actionable* they agree with each other at 0.76 and with the human at 0.06, and three quarters of their consensus is reproducible from eight surface features. Refitting on that would optimise

surface form — the length collapse just measured. Only **correct** and **targeted** are not surface-driven, and only those two are safe to average.

Low per-turn agreement also has a benign cause: little room to disagree. **actionable** is rated 3 on 74% of turns by the models and 65% by the human, so everyone is choosing between 2 and 3. The score *distributions* match on all six qualities, differing only in which turn gets which — the regime where averaged model ratings are trustworthy for comparing policies and untrustworthy for labelling single turns.

7 Ablation Results

A rebuilt rubric, a measured learning rate, two failed repairs, and a ceiling belonging to the reward model.

The blind evaluation left three things to repair: one clear success, one measurement that changed a default, and two failures worth reporting as failures.

7.1 The rubric was measuring fewer things than it claimed

Three properties of the original labels, none arguable:

- **actionable** and **elicits** correlate at **0.95**: one quality with two names, counted twice.
- The five qualities have effective rank **2.9**; the first principal component explains 47% of the variance.
- Nine turns are top-quartile on **leak**, **targeted**, **actionable** *and* **elicits** at once, median length **12 words** against a corpus median of 30.

That last item is the length collapse, visible in the labels before any training. A short bare question satisfies four of five qualities simultaneously, so the reward had a degenerate optimum and RL found it.

The cause is structural: every original quality measures what the tutor *withheld* — didn’t reveal, didn’t over-explain, left the thinking to the student — and none measures what it *contributed*. A rubric of withholding measures is maximised by maximal withholding. Across 70 effect sizes, elaborated feedback is $d = 0.49$, knowledge of the correct response $d = 0.32$, bare knowledge of results $d = 0.05$. The reward was maximising the $d = 0.05$ end.

7.2 The replacement, and how much better raters agree on it

Six qualities replace the six: **guidance** (does the turn supply anything usable), **locates** (does it point at something specific in the student’s message), **hands_over** (does it hand over the next move and name it), **verdict** (does its right/wrong signal match what the student said), **leak**, **correct**. A seventh, **student_state**, is rated on the *student’s* message and never enters the reward; it exists so contingency can be computed later. Five models rated 44 turns, one full pass per quality so judging one could not colour the next.

Two results beat the averages. **hands_over** scores 0.81 where **elicits** scored 0.56, and the only change is framing: “does the turn hand over the next move” rather than “how much thinking does it ask for.” The literature predicted this — behavioural framings of the construct reach κ 0.55–0.95, cognitive-demand framings 0.09–0.34. And **correct** went 0.18 $\rightarrow$ 0.80 through an interface change, not a rewording: raters had been re-deriving the mathematics, and are now shown a generated worked solution to check against.

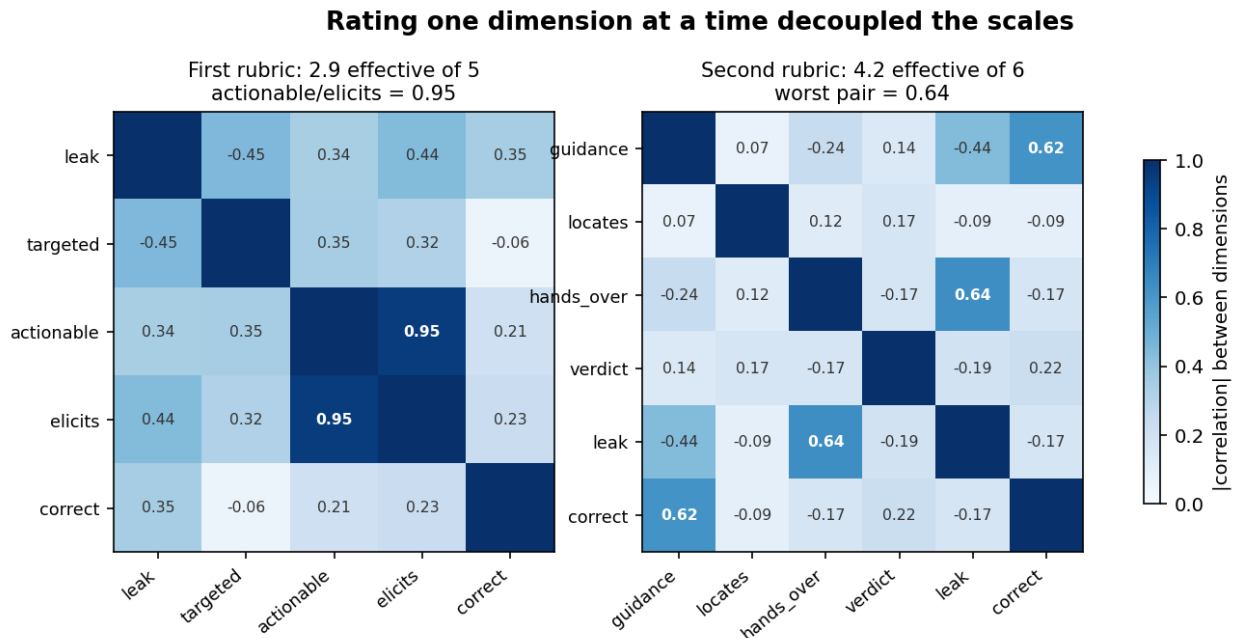


Figure 7: Absolute correlations between qualities. Rating one quality at a time across all turns, rather than six per turn, breaks the coupling that made actionable and elicits one measurement.

Quality	Weighted κ	Ceiling	Replaces
leak	0.83	0.93	same wording; was 0.51
hands_over	0.81	0.93	elicits, which was 0.56
correct	0.80	0.93	same construct, reference solution now shown; was 0.18
verdict	0.73	0.86	new
locates	0.67	0.83	targeted , which was 0.57
guidance	0.51	0.74	new, and the weakest

Table 7: Inter-rater agreement across five raters. All six clear the 0.4 threshold; two of the original six did not.

7.3 The reward is largely a length penalty, and the new rubric does not fix that

Agreement was the wrong property to check alone. What the dimensions correlate *with* explains more of this project’s history than what raters agree on. Running the head over all 600 labelled turns, against the seven-rater consensus on the same turns:

Applying the reward signs resolves the picture. **leak** is negated, so its +0.64 becomes −0.64 and joins the other three: **correct** −0.58, **elicits** −0.52, **actionable** −0.42, with **targeted** alone at +0.03. **The composite reward correlates with length at −0.60 in the probe and −0.57 in the labels.**

That is not a side effect but most of what the reward is, and it accounts for the sequence rather than sitting alongside it: why arm A collapsed to 12 words, why a length term had to be introduced, why that term supplies roughly 70% of the measured gain — it is the only one pushing back — and why the probe dimensions collected only 40% of their available headroom, the remainder lying behind the collapse the band blocks. **targeted** is the exception, and it is also where surface features

	Probe	Labels
actionable $\leftrightarrow$ elicits	0.96	0.95
leak $\leftrightarrow$ log(words)	+0.64	+0.53
correct $\leftrightarrow$ log(words)	−0.58	−0.43
elicits $\leftrightarrow$ log(words)	−0.52	−0.50
actionable $\leftrightarrow$ log(words)	−0.42	−0.40
targeted $\leftrightarrow$ log(words)	+0.03	+0.02
mean $ r $ among the five	0.43	0.37
first principal component	51%	47%
effective rank	3.07 / 5	3.31 / 5

Table 8: The probe amplifies the label structure and invents none of it: every correlation matches the labels in sign and exceeds it by 0.01–0.15. The redundancy is a property of the ratings.

do worst (0.36 against 0.85); it is the one dimension length cannot fake and the one the policy improved least on.

V2 fixes the redundancy and not the length coupling. guidance was included partly as a length-*positive* counterweight, so this is checkable: nine raters scored 44 turns on all six V2 dimensions. But 24 of those 44 are manufactured negatives, each corrupted along a single dimension, which decorrelates the dimensions by construction and says nothing about the rubric. Excluding them leaves 20 real turns.

	V1 ($n=600$)	V2, all 44	V2, real only ($n=20$)
mean $ r $	0.37	0.22	0.28
max $ r $	0.95	0.65	0.70
first PC	47%	36%	40%
effective rank	3.31 / 5	4.88 / 6	4.53 / 6
composite vs log(words)	−0.57	−0.37	−0.62

Table 9: The last column is the one to read. V2 genuinely removes the redundancy — the 0.95 duplicate is gone and effective rank rises from 3.31 of 5 to 4.53 of 6 — while the length coupling is unchanged at −0.62 against V1’s −0.57.

Per dimension on real turns: correct −0.66, verdict −0.53, hands_over −0.51, leak −0.26, guidance −0.12, locates −0.06. The two dimensions designed to be length-neutral are, and the other four swamp them; guidance was meant to be length-positive and came out neutral. At $n = 20$ the interval on −0.62 is roughly $[-0.84, -0.24]$, so this cannot show V2 is worse — only that it does not help.

The conclusion is that rubric wording is the wrong lever here. Decorrelating the dimensions from each other and decorrelating them from length are separate problems; V2 solves the first and leaves the second untouched, because raters score shorter turns higher on most pedagogical qualities under either rubric. Brevity is confounded with quality in their own judgements, so any head fitted on this corpus inherits it. The lever is the corpus — turns in which length and quality vary independently — which has now been identified as necessary twice and still does not exist.

7.4 Negative examples had to be manufactured

The pool contained almost no bad turns: in a 20-turn pilot *locates* never received the bottom score and *guidance* received it once, and a head cannot learn an anchor from zero examples. Real turns were corrupted along one quality each — the trick that took answer-injection from 93% to 3% — and rated blind. Four of six recipes landed unanimously; *guidance* and *locates* landed half the time, because “strip out the content” can be satisfied halfway. Restated as prohibitions — not one number or content word from the student’s message may appear — overlap with the student’s own words fell from 5.6 to 0.2 for *locates*.

A methodological failure worth recording. The corrupted turns’ identifiers encoded which quality had been corrupted. One rater read the answer off the identifier prefix rather than judging the turn, and said so in its report; its labels were discarded and identifiers are now salted hashes. It surfaced only because that rater described its method honestly, which is a thin thing to have relied on.

7.5 The learning rate, measured rather than chosen

Four 35-step runs, identical apart from the rate, compared on reward gained per unit of KL drift. Reward alone ranks runs by learning rate and is uninformative; the ratio has an interior maximum.

Rate	Reward gain	KL	Gain/KL	Words	In band
2e-5	+0.526	0.074	7.11	50→40	0.90
1e-5	+0.348	0.080	4.35	50→31	0.81
4e-5	+0.526	0.122	4.31	50→43	0.92
8e-5	+0.606	0.172	3.52	49→36	0.94

Table 10: 2e-5 buys the same gain as 4e-5 for 40% less drift.

The word-count column is the more interesting result: the *slowest* rate produced the *shortest* turns, 31 words against 40, and the in-band share rose monotonically with the rate. At 1e-5 the probe’s pull toward brevity still dominates and the policy sits at the band’s lower edge; from 2e-5 up it reaches the band and stays.

7.6 Two attempts to decouple leak from length, both unsuccessful

The probe ties leakage to length at +0.70 on policy-generated turns against the human’s +0.43, and that distortion made both automated judges rank arm C below arm A when the human preferred it 29–11.

The second attempt is the informative one. Residualising length out of the states should leave the head no length direction to rediscover, and in-distribution it does. Out of distribution the coupling *rose* to +0.74, the ridge penalty collapsed to its smallest value, and prediction spread grew $0.46 \rightarrow 0.64$: the fit had begun memorising.

The diagnosis: in-distribution the head’s coupling is 0.44 against the labels’ 0.37, so there is almost nothing to correct there. The +0.70 is distribution shift — on policy-generated text the head has never seen, length becomes a far stronger cue — and no correction estimated in-distribution reaches a problem that exists only out of it. Fixing it requires labels on policy-generated turns.

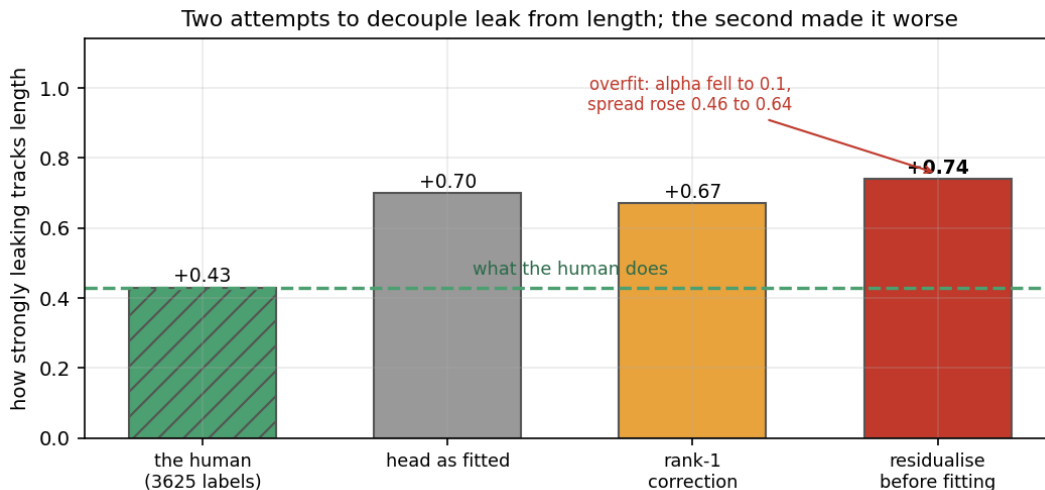


Figure 8: A rank-1 correction toward the labels’ own coupling achieved little; residualising length out of both states and target before fitting made it worse.

Until then the explicit length term is the only predictable lever, so it went from a hard band to a band with linear ramps: a hard band scores 59 and 200 words identically, so nothing pushes a runaway turn back.

Designing the length reward

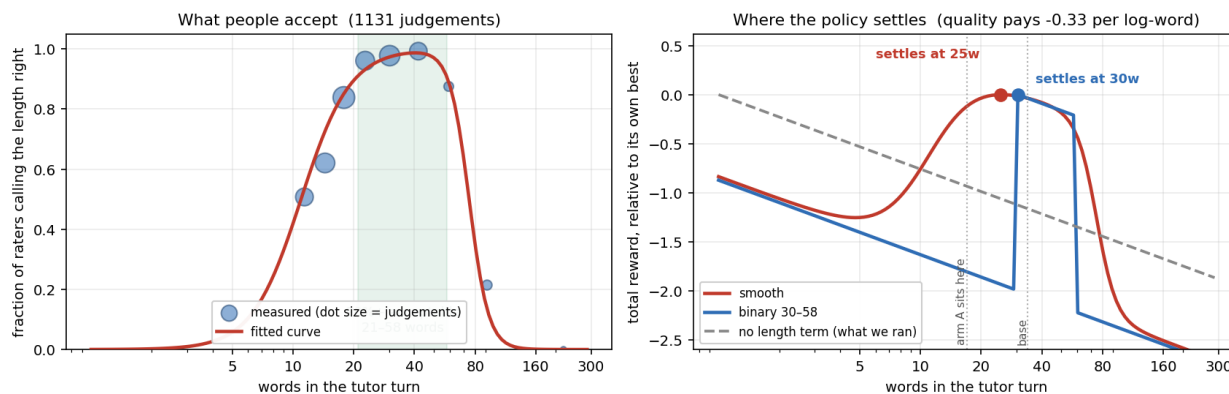


Figure 9: Left: acceptable lengths, from 1131 human judgements. Right: where the policy settles once the probe’s own pull toward brevity is accounted for. The band’s interior is flat because that is what fixes the resting length; a smooth peak drifts as the pull changes during training.

7.7 The learning-rate ladder

Three runs at $2e-5$, $4e-5$ and $8e-5$, identical otherwise — same probe, same length term, same 200 steps. This is the comparison arm B could not give, since z-scoring shrank its reward scale and thereby raised its effective step size $3.4\times$ as a side effect.

All three converge to the same reward and the faster ones only arrive sooner — a different answer from the one the question presupposed. The reward is not a quantity a larger step can push higher: its ceiling is what the probe can express, and all three optimisers reach it. Arm E’s

Arm	Rate	Held-out reward	Words	KL, final	KL, peak	Steps to 3.80
E	2e-5	3.84	27.2	0.25	0.31	133
F	4e-5	3.88	27.7	0.29	0.41	69
G	8e-5	3.86	28.2	0.29	0.41	65

Table 11: Four times the learning rate buys 0.02 more reward and arrives twice as fast. Reward and length are on held-out questions; KL is the mean over the last ten steps and its peak over the run; the last column is the first step whose nine-step mean reaches 3.80.

trajectory agrees: reward rose $3.13 \rightarrow 3.84$ over 200 steps, with successive 20-step gains of $+0.31$, $+0.14$, $+0.13$, $+0.03$, $+0.02$, then eight increments of 0.03 or less.

The drift saturates rather than scaling with the rate. $2e-5 \rightarrow 4e-5$ costs 0.04 nats; doubling again costs nothing, with F and G ending at the same 0.29 and peaking at the same 0.41. Past $4e-5$ the extra step size buys arrival time, not distance travelled, which makes $4e-5$ better supported than the $2e-5$ the gain-per-KL measurement selected — that ran 35 steps, before either arm had converged.

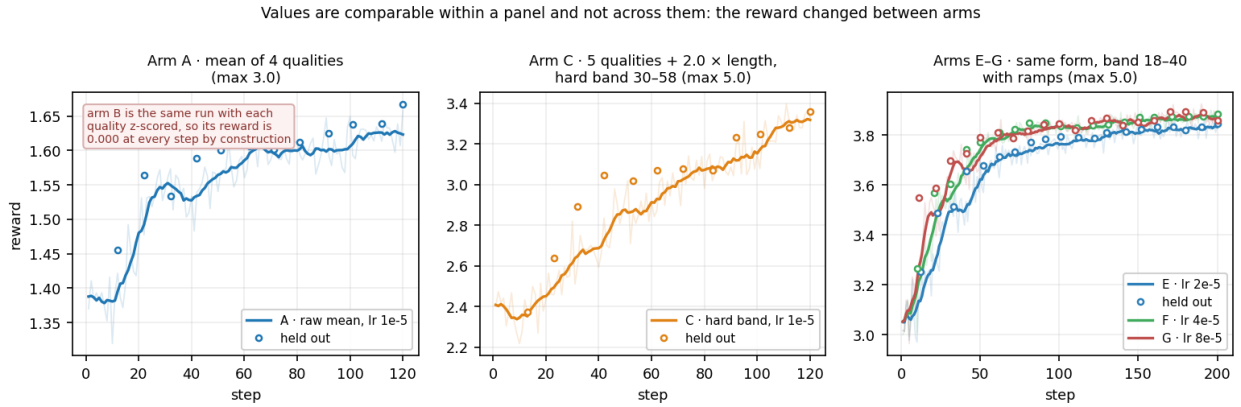


Figure 10: Reward over training, one panel per reward *definition*. Values are comparable within a panel and not across them: arm A averages four qualities, arm C five plus a length term, and E–G use arm C’s form with a different band. Open circles are held-out questions, tracking the training curve throughout.

What this settles about arm B. Arm B claimed $+0.49$ and was paid $+0.37$ at $3.4\times$ arm A’s effective step size. Arms F and G are $2\times$ and $4\times$ arm E’s and post the same reward, so arm B’s over-claiming was not a generic consequence of larger steps — the scalarisation contributed something step size alone does not. That question deserved, and never got, a test at matched step size.

Where the remaining headroom is. Arm E ends at 3.84 against a maximum of 4.20. The length term is saturated at 1.96 of 2.00 with 98% of turns in band, while the five pedagogical qualities sit at 1.88 of 2.20. So roughly 85% of what remains is in the qualities, and leak is furthest from its ceiling — the same quality whose length coupling two repairs failed to remove.

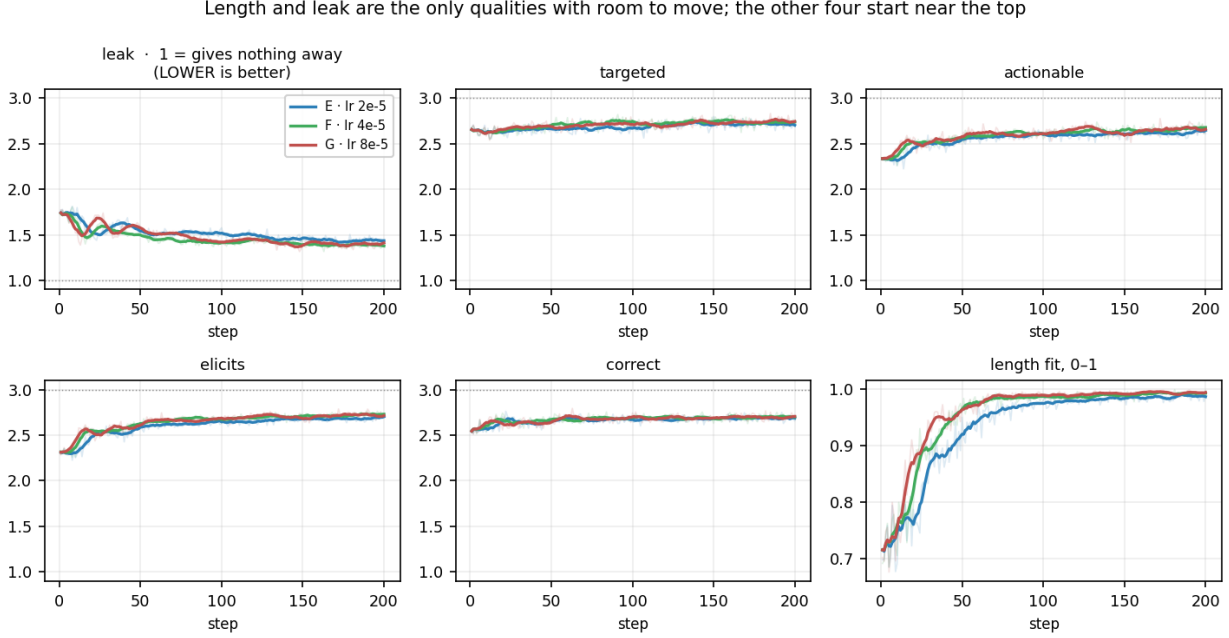


Figure 11: Which qualities actually moved. Only length and leak have real room: targeted, actionable, elicits and correct start near 2.6 of 3 and finish within 0.15 of where they began. This is Table 11’s ceiling seen per-quality.

7.8 The ladder rated blind

Sixty held-out questions answered by all four policies — untrained plus three rates — giving 240 turns, blinded and rated by five calibrated models.

Quality	E (2e-5)	F (4e-5)	G (8e-5)
elicits	+0.48 (73%)	+0.52 (78%)	+0.49 (75%)
actionable	+0.43 (72%)	+0.41 (66%)	+0.39 (66%)
leak	+0.33 (67%)	+0.32 (70%)	+0.31 (64%)
correct	+0.19 (66%)	+0.17 (68%)	+0.13 (62%)
length_fit	+0.18 (66%)	+0.18 (67%)	+0.17 (66%)
targeted	+0.14 (69%)	+0.19 (69%)	+0.13 (68%)
total	+0.35 (80%)	+0.36 (86%)	+0.33 (80%)

Table 12: Gain over untrained, paired by question. Percentages are the share of questions where the trained turn was rated better. Every gain exceeds its margin of error.

Every arm beats base on every quality, which no earlier arm managed. Arm C won on four and lost leak because getting longer cost it there; arm E wins on all six, withholding more than base *while* being the right length — unreachable earlier, when the only lever on length was a probe that penalises it through leak.

The length problem is closed: 100% of turns rated right-length against base’s 8% / 82% / 10%. Arm C’s hard band left 29% too short; moving the band to 18–40 words and adding ramps removed both tails.

The three rates are interchangeable. Totals of 1.73, 1.74 and 1.71, every quality within

The learning-rate ladder is healthy at every rate, and differs only in scale

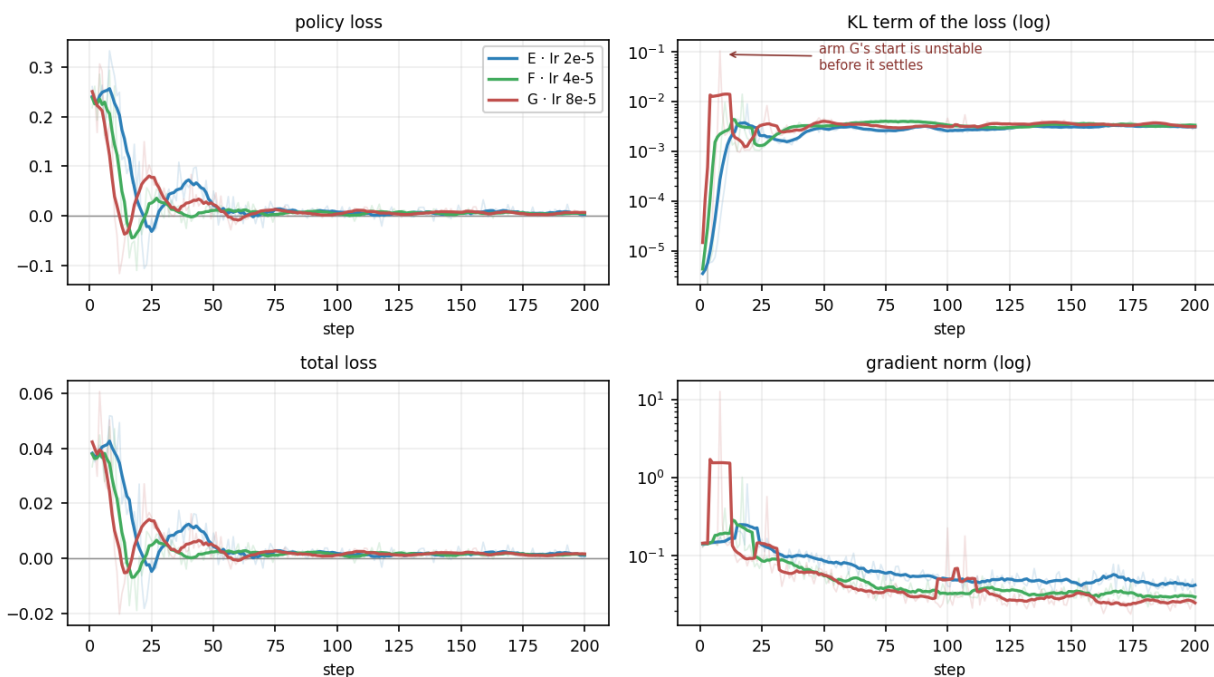


Figure 12: Well-behaved at all three rates. Both right-hand panels are logarithmic because arm G spikes in its first ten steps — gradient norm 12.8 and KL term 0.106 against medians of 0.033 and 0.0034 — the only sign 8e-5 starts unstably before settling. Arm F’s curve is stitched from three W&B runs, since the partition preempted it twice and each requeue resumed from the last checkpoint.

noise. With reward converging to 3.84–3.88 and drift rising $0.25 \rightarrow 0.29$ then flattening, this is the cleanest statement the project can make about the optimiser: it chooses how fast, the reward model chooses how far.

Why this should not be believed yet. These are model raters, and on arm C they were confidently wrong: they tie leakage to length at +0.59 against the human’s +0.43, and that bias made them rank arm C below arm A when she preferred it 29–11. Arm E is *shorter* than arm C, so the bias now runs in its favour — exactly when it should be trusted least. A blinded human preference test over 48 pairs is outstanding and decides whether Table 12 is a result or an artefact.

7.9 All seven arms on one axis

The reward changed definition four times, so no reward column compares all seven arms. But the four *original* qualities — leak, targeted, actionable, elicits — were scored by the same four heads in every run, so their mean is comparable throughout.

Two things only this table shows. **Arm B’s drift was $3.6\times$ arm A’s** (0.567 against 0.158) at the same nominal rate, which puts a number on the confound: z-scoring shrank the reward scale, and with centred advantages that raises the effective step size rather than reweighting the qualities. Arm B was never a test of scalarisation. And **arm C gained least of any completed arm** (+0.116 against +0.26 to +0.34), because it spent its capacity getting longer — the same trade that cost it leak and made the probe rank it below arm A while the human ranked it above.

Higher rates travel further from the starting policy for the same reward — the cost of the ladder

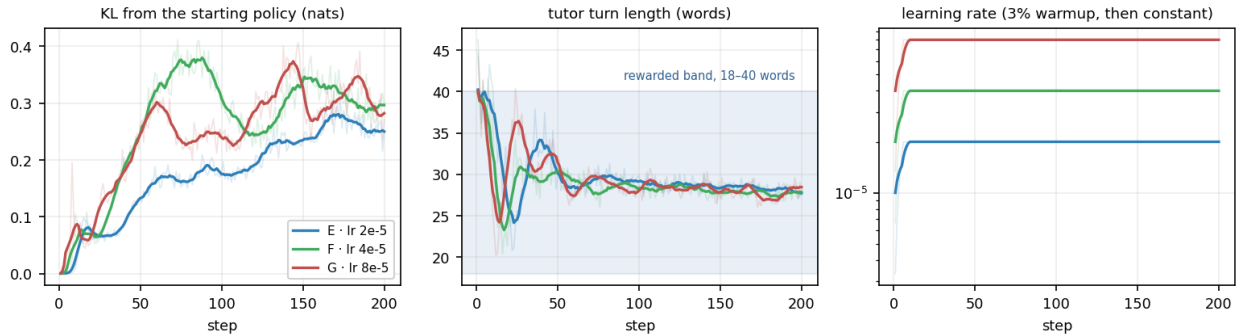


Figure 13: Drift, generated length, and the schedule. Length falls *into* the rewarded band and stops, which is what the ramps were built to do and what arm C’s hard band failed at. Four diagnostics are constant and so not plotted: clipped tokens exactly 0, importance-ratio variance 10^{-15} , mean advantage 10^{-17} , and no group of 16 samples ever scoring uniformly — all consequences of strictly on-policy single-epoch updates, where the importance ratio is 1 by construction.

Arm	Rate	4-quality mean, start	4-quality mean, end	Gain	KL	Words
A	1e-5	1.370	1.627	+0.257	0.158	38 → 27
B	1e-5, z-scored	1.388	1.724	+0.336	0.567	43 → 23
C	1e-5, band 30–58	1.386	1.502	+0.116	0.184	46 → 39
D	1e-5, ramps	—	—	—	—	—
E	2e-5, band 18–40	1.387	1.650	+0.263	0.250	46 → 28
F	4e-5	1.382	1.685	+0.303	0.293	45 → 28
G	8e-5	1.375	1.677	+0.302	0.285	44 → 28

Table 13: Every arm on the one quantity that never changed definition, averaged over the last ten steps as scored by the probe — what the probe believes, not what a human paid. Arm D is blank because the partition preempted it at step 46, before its first checkpoint.

8 Capability Retention

No detectable loss on ARC-Challenge, MMLU stem or GSM8K: −0.3 points pooled, bounded at −1.5.

Everything so far compared tutoring against tutoring, but the policy was optimised for 200 steps against a reward built entirely from the shape of a turn, which says nothing about whether it can still do arithmetic or recall a fact — and narrow objectives are documented to cost capability elsewhere.

So both trained arms and base were asked to *answer* rather than *teach*, on three benchmarks none of them trained on. Multiple-choice items are scored by the loglikelihood assigned to each option letter, which avoids depending on format compliance; GSM8K is generated and the final number parsed. Every arm sees an identical 400 items per benchmark, decoding is greedy, and differences use McNemar’s exact test on the items where two models disagree.

Nothing moved. Pooled over 1200 items, arm E is 0.3 points below base and arm G 0.4, both inside noise. On ARC-Challenge arm E is not merely equal to base but identical item for item: none of the 400 changed hands.

	base	arm E (2e-5)	arm G (8e-5)
ARC-Challenge	70.8%	70.8%	70.8%
MMLU stem	42.5%	42.0%	42.0%
GSM8K	83.8%	83.2%	83.0%
pooled change	—	−0.3 [−1.1, +0.5]	−0.4 [−1.5, +0.7]

Table 14: Capability is unchanged. Pooled change is the paired difference over all 1200 items in percentage points, with a 95% interval; $p = 0.54$ and $p = 0.56$.

The null is bounded, not merely unrefuted. This matters more than the p -value, since “we observed no change” and “we could not have observed a change” read identically. Pooled, we exclude any drop worse than roughly 1.5 points. Per benchmark the intervals are much wider — GSM8K’s reaches -3.6 — which is why the pooled figure carries the claim. It is also the expected outcome: arm E ends at 0.25 nats, and a rank-32 adapter over a frozen 7B has limited capacity to overwrite what the base knows. Had capability fallen anyway, the tutoring gains would have been paid for out of something else.

Two harness faults preceded these numbers, both giving confident wrong answers. Generation was first capped at 24 tokens, on the assumption a direct question draws a direct answer; instruction-tuned models reason first, so half the GSM8K generations were truncated mid-derivation and measured accuracy came out at 1.8% against a published $\sim 67\%$. Raising the budget to 512 still left 30% of MMLU items unparseable — and that failure is correlated with the quantity under test, since a more verbose policy loses more items to truncation. Under that harness arm G appeared to fall $41.0\% \rightarrow 36.2\%$, where option loglikelihoods give 42.5% against 42.0%. A parse-failure rate that differs across arms is not noise but a bias with a sign, and it pointed toward concluding the trained models were worse.

9 Limitations

- **Questions were held out from the policy but not from the probe.** The 250 training and 50 evaluation questions do not overlap, but 45 of the 50 were among those the probe was fitted on. Nothing leaks, since the probe is frozen, but any agreement measured here flatters it.
- **One human, 17 questions rated and 28 pairs decided.** Every claim about which policy is better rests on 60 questions rated by models whose per-turn agreement with her is weak, plus her own 17. They agree, which is why either is believable, but neither is a large sample.
- **Only the final checkpoint was evaluated.** Checkpoints exist every 10 steps, so if quality peaked earlier while reward kept climbing, this would not show it.
- **The probe reads text far from what it was fitted on.** The encoder is frozen, but it was fitted on base-model turns and by step 200 reads policy text at 0.25 nats and 28 words against 46. That shift defeated both leak repairs.
- **Why nobody agrees per turn on elicits and actionable is unexplained.** Both are near-saturated here, so what remains may be taste rather than a learnable signal.

Academic ability after 200 steps of tutoring-shaped RL

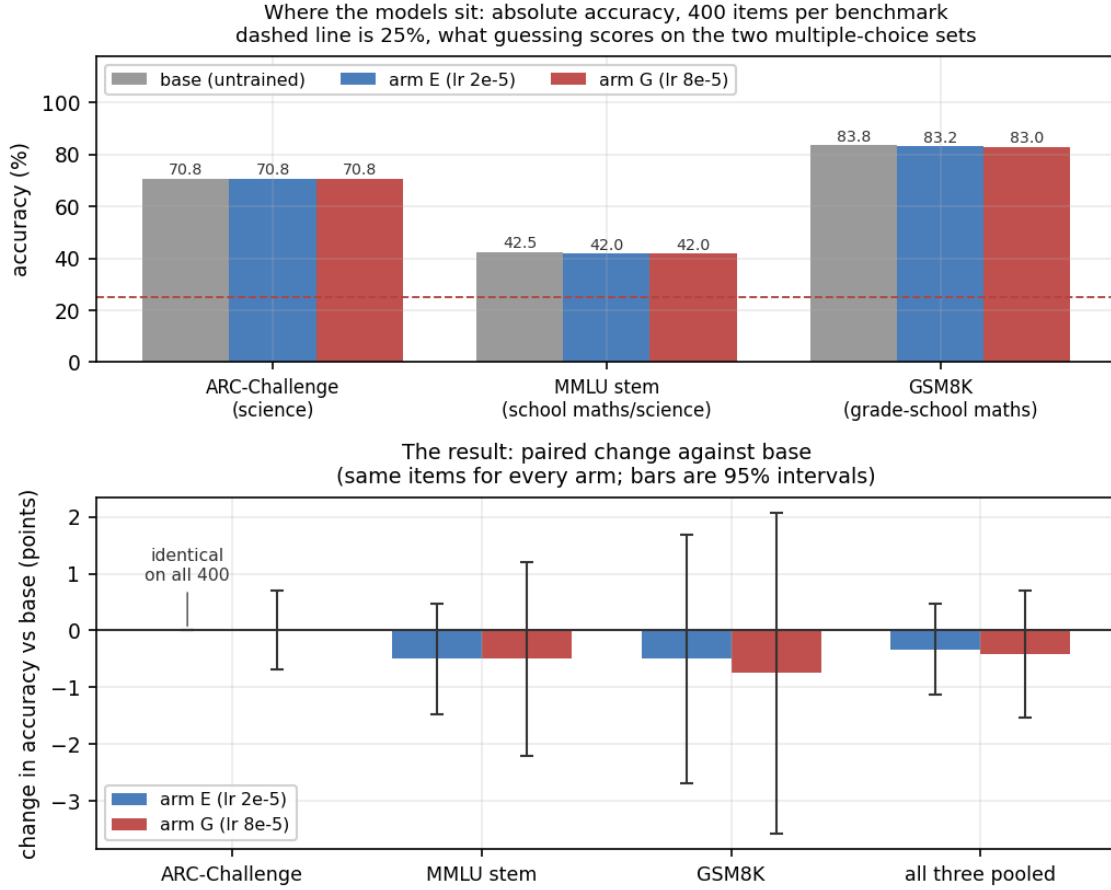


Figure 14: Top: absolute accuracy, which situates the result — MMLU stem at 42.5% is close enough to the 25% chance floor that little of the scale is in play, while GSM8K at 83.8% has little headroom, and a 0.3-point change means something different in each case. Bottom: the paired difference, plotted separately because at 42% and 84% the raw bars are visually identical whatever the gap.

10 Conclusion

A ridge fit on a frozen model’s activations is a usable RL reward for tutoring quality, and its claimed improvement held up under a blind check by raters who could not see it — which the predecessor outcome-based reward conspicuously failed. It is not a fine instrument: it cannot separate two policies that have both climbed it, and optimising it harder produced a policy that claimed more than it was paid.

What we did not expect is that optimising it would be the easy part. Three learning rates spanning $4\times$ reach the same place, four of six qualities sit near their ceiling before training starts, and the one quality with real headroom is the one whose length coupling resisted two principled repairs. The optimiser was never the constraint. Three things follow:

1. **Collect labels on policy-generated turns.** The $+0.70$ length coupling exists only out of distribution, so no correction estimated on the existing corpus reaches it. This is the highest-value annotation left and it unblocks the only quality with headroom.
2. **Use 4e-5, not 2e-5.** Same reward in half the steps at the same final drift. The gain-per-KL

measurement that chose $2e-5$ ran 35 steps, before either arm converged.

3. **Evaluate intermediate checkpoints.** If quality peaks before reward does, that is where to stop, and locating it beats a run that happened to stop early.